An example of sample size determination in the 2AFC study

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Summary

This example shows the process of determining an appropriate sample size for a two-alternative forced choice (2AFC) study aimed at comparing the performance of a deep learning-based model observer (DLMO) with that of human observers on the Rayleigh task. By specifying the desired significance level and the expected difference in performance (e.g., proportion correct) between two types of observers, we explore different combinations of cases, readers, and study designs (fully crossed or split-plot design) needed to detect the effect with sufficient confidence.

Basic Settings

Test Data and Task

Test samples in this example consisted of synthetic objects with artificially inserted signals. The background was generated using a stochastic object model that had been trained on clinical MR imaging data. Each synthetic object contained either a singlet (a Gaussian-convoluted rod) or doublet signal (two Gaussian-convoluted dots) that was inserted into the white and gray matter regions of the brain. The separation of two dots in doublet signals and the length of rod in singlet signals ranged from 5 mm to 9 mm. Examples of the background and two types of signals are shown in Fig. 1.

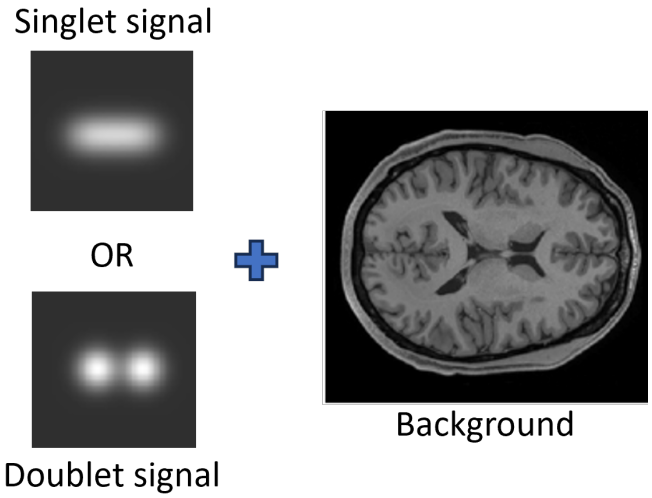


Figure 1: Examples of background and two types of signals.

These objects were scanned by a simulated MR system at two different acceleration rates (x4 and x8). Images were then reconstructed by two methods (physic-based rSOS and Unet). Thus, there will be $2 \times 2 = 4$ reader

studies in the pivotal study. Examples of reconstructed MR images are shown in Fig. 2.

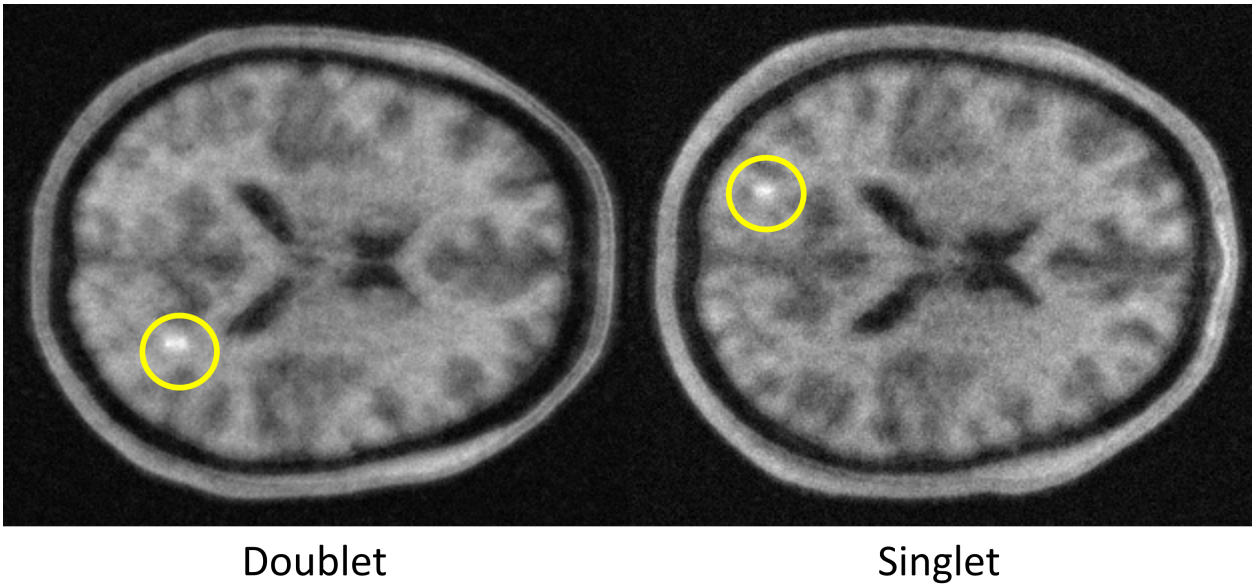


Figure 2: Examples of reconstructed MR images with singlet (right) and doublet (left) signals. These examples were reconstructed by rSOS at acceleration x4. Yellow circles indicate signal locations.

In the 2AFC study, pairs of images are presented to readers side by side, with one image containing a singlet signal and the other containing a doublet signal. Readers are instructed to identify which image contains the doublet signal.

Hereafter, a pair of test images (or a trial in the 2AFC study) is referred to as one **case**, denoted by C in the subscripts.

Objective

The objective is to investigate whether a deep learning-based model observer (DLMO) performs similarly to human observers on the Rayleigh discrimination task.

Denote human observer proportion correct by θ_0 and DLMO proportion correct by θ_1. Also, denote similarity margin by ΔM. We have

$$H_0 : |\theta_0 - \theta_1| \geq \Delta M,$$
$$H_1 : |\theta_0 - \theta_1| < \Delta M.$$

Pilot study

The process to determine sample size begins with a pilot study, which provides parameter estimates of the difference in proportion correct (PC) and its variance as a starting point. The pilot study involved three human readers. These observers were presented with 50 cases (50 trials or $N_C = 50$) in a 2AFC study. The pilot study was fully crossed, which means all cases were read by all readers.

The DLMO was trained on 80,000 pairs of images with singlet and doublet signals and tested on a test dataset with $N_C = 4000$ cases. For the purposes of this document, the details will not be presented. Using the iMRMC application (FDA/CDRH 2022), we calculated the PC achieved by the DLMO and its variance, denoted by θ_1 and V_1, respectively, as follows.

$$\theta_1 = 0.88,$$
$$V_1 = 0.0001188201,$$

where V_1 is the variance of PC achieved by DLMO.

Variance Estimation using iMRMC

First, we estimate the variance of PC achieved by human readers in the pilot study.

```
# Load in the iMRMC package
library(iMRMC)

# Load in the pilot data
csv_fname_ho <- "pilot_study_curated.csv"
data_ho <- read.csv(csv_fname_ho,  header = TRUE)

# Analyze the pilot data use this function.
# Note: The pilot data was initially binary but has been curated so that can be used in iMRMC.
# More info: https://github.com/DIDSR/iMRMC/wiki/MRMC-analysis-of-binary-data
results <- doIMRMC(data_ho)

# Variance of PC achieved by human readers
V_0 <- results$MLEstat$varAUCA
cat("Variance of PC achieved by human readers in the pilot study: ",V_0,sep = "")
```

```
## Variance of PC achieved by human readers in the pilot study: 0.0009179259
```

This variance can be decomposed into several components. There are multiple ways of the decomposition and here we implement the BDG decomposition proposed by Gallas et al.(Gallas et al. 2009; Gallas 2006; Gallas and Brown 2008; Gallas, Pennello, and Myers 2007). The variance of PC achieved by human readers is decomposed as a scalar product between the coefficients, denoted by vector $\mathbf{c}$, and the moments, denoted by vector $\mathbf{M}$.

$$V_0 = c_1 M_1 + c_4 M_4 + c_5 M_5 + c_8 M_8.$$

The moments $\mathbf{M}$ can be estimated by the iMRMC.

```
bdg_comp <- results[["varDecomp"]][["BDG"]][["MLE"]][["comp"]]
M1 <- as.numeric(bdg_comp[1, "M1"])
M4 <- as.numeric(bdg_comp[1, "M4"])
M5 <- as.numeric(bdg_comp[1, "M5"])
M8 <- as.numeric(bdg_comp[1, "M8"])
```

M_1	M_4	M_5	M_8
0.9333	0.8712	0.9067	0.8711

Also, we calculate PC achieved by human readers in the pilot study.

```
theta_0 <- results$MLEstat$AUCA
cat("PC achieved by human readers in the pilot study: ",theta_0,sep = "")
```

```
## PC achieved by human readers in the pilot study: 0.9333333
```

Since samples to test the DLMO were independent from those to test human readers, and DLMO is independent from human readers, the standard error of difference in PCs between human readers and DLMO is given by

$$SE_{diff} = \sqrt{V_0 + V_1}$$

Sizing

We targeted a standard error of 0.0155, believing that would allow ample precision for detecting a similarity between DLMO and human readers within a margin of 0.1 at a significance level of 0.0125 (0.05 split evenly among 4 reader studies). The difference between PCs from DLMO and human readers is assumed to be 0.053, as observed in the pilot study. We will show two types of designs for the pivotal study, including the fully crossed design and the split-plot design.

Fully crossed design

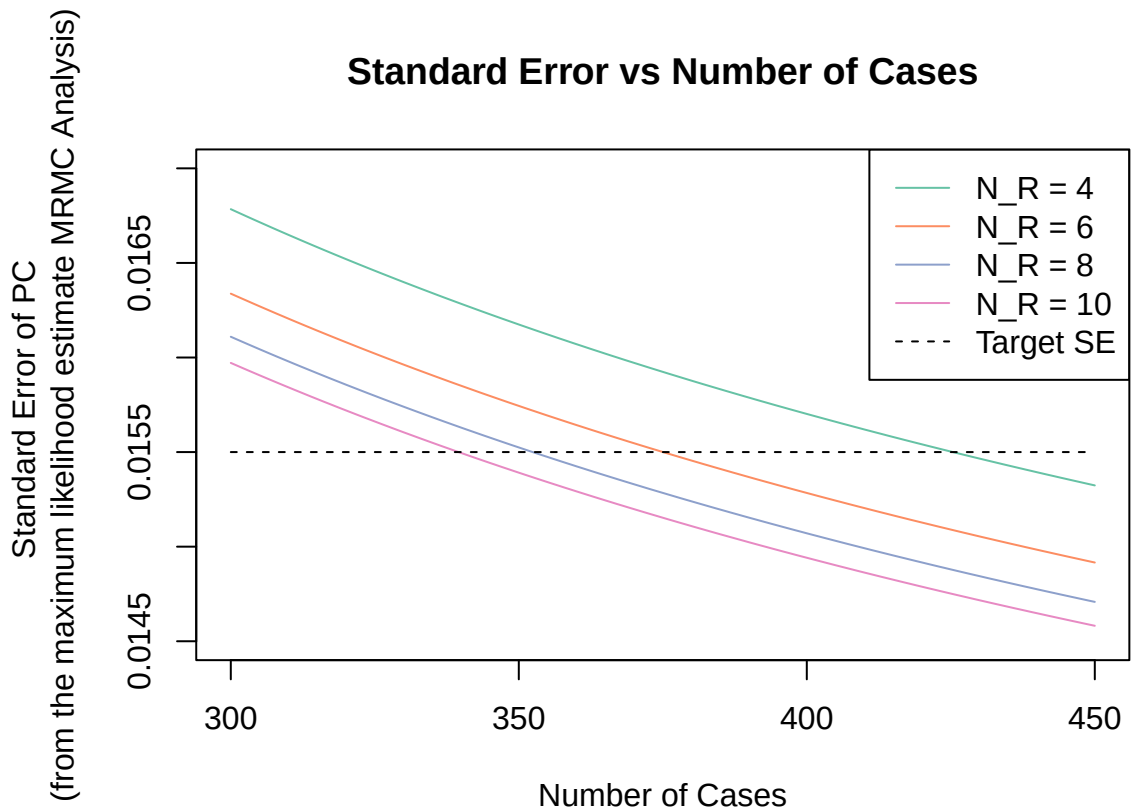
In the fully crossed design, all cases are read by all readers. The coefficients of moments of variance in a fully crossed 2AFC study are as follows (Gallas, Pennello, and Myers 2007).

c_1	c_4	c_5	c_8
$1/N_C N_R$	$(N_C - 1)/N_R N_C$	$N_R - 1/N_R N_C$	$[(N_C - 1)(N_R - 1) - N_R N_C]/N_R N_C$

By changing the number of cases and the number of readers, we plot the standard error of difference in PCs between DLMO and human readers.

```
# Set N_C range
N_C_start <- 300
N_C_end <- 450

# Set N_R range
N_R_start <- 4
N_R_end <- 10

# Create an empty figure
op <- par(mar=c(5, 6, 4, 2) + 0.1)
plot(1, type = "n",
     xlim = c(N_C_start, N_C_end),
     ylim = c(0.0145, 0.017),
     xlab = "Number of Cases",
     ylab = "Standard Error of PC \n(from the maximum likelihood estimate MRMC Analysis)",
     main = "Standard Error vs Number of Cases")
par(op)

# Define line colors
library(RColorBrewer)
colors <- brewer.pal(50, "Set2")

# First loop over reader numbers
R_index <- 1
for (N_R in seq(N_R_start, N_R_end, by = 2)){
  se_BDG_array_fc_r <-
```

```r
    rep(0, 1) # array using replicate command

  # With N_R readers, loop over case numbers
  index <- 1
  for (N_C in N_C_start:N_C_end) {
    # ---- Compute BDG coefficients for fully crossed ----
    c1fc <- 1 / (N_R * N_C)
    c4fc <- (N_C - 1) / (N_R * N_C)
    c5fc <- (N_R - 1) / (N_R * N_C)
    c8fc <- ((N_C - 1) * (N_R - 1) - (N_R * N_C)) / (N_R * N_C)

    var_ho_fc <- c1fc * M1 + c4fc * M4 + c5fc * M5 + c8fc * M8
    se_BDG_array_fc_r[index] <- (var_ho_fc + V_1) ^ .5

    index <- index + 1

  }

  # Plot standard error with N_R readers and N_C cases
  lines(
    N_C_start:N_C_end,
    se_BDG_array_fc_r,
    col = colors[R_index],
    type = "l",
  )
  R_index <- R_index + 1
}

# Plot the target standard error
lines(N_C_start:N_C_end,
      rep(0.0155, index - 1) ,
      lty = 2,
      col = "black")

legend(
  "topright",
  legend = c("N_R = 4","N_R = 6","N_R = 8","N_R = 10", "Target SE"),
  col = c(colors[1:4], "black"),
  lty = c(1,1,1,1,2)
)
```



We observe that standard error with $N_R = 4$ falls below the target standard error after approximately $N_C = 430$. This reveals that *for the considered margin, to show the similarity of DLMO to human readers with a significance level of 0.0125,* ***4*** *readers need to read at least* ***430*** *cases in a fully crossed 2AFC study.*

Split-plot design

In the split-plot design, readers do not read the entire set of test samples but instead read only a subset. There are multiple designs, and we will show two types of them here.

Flat without overlap

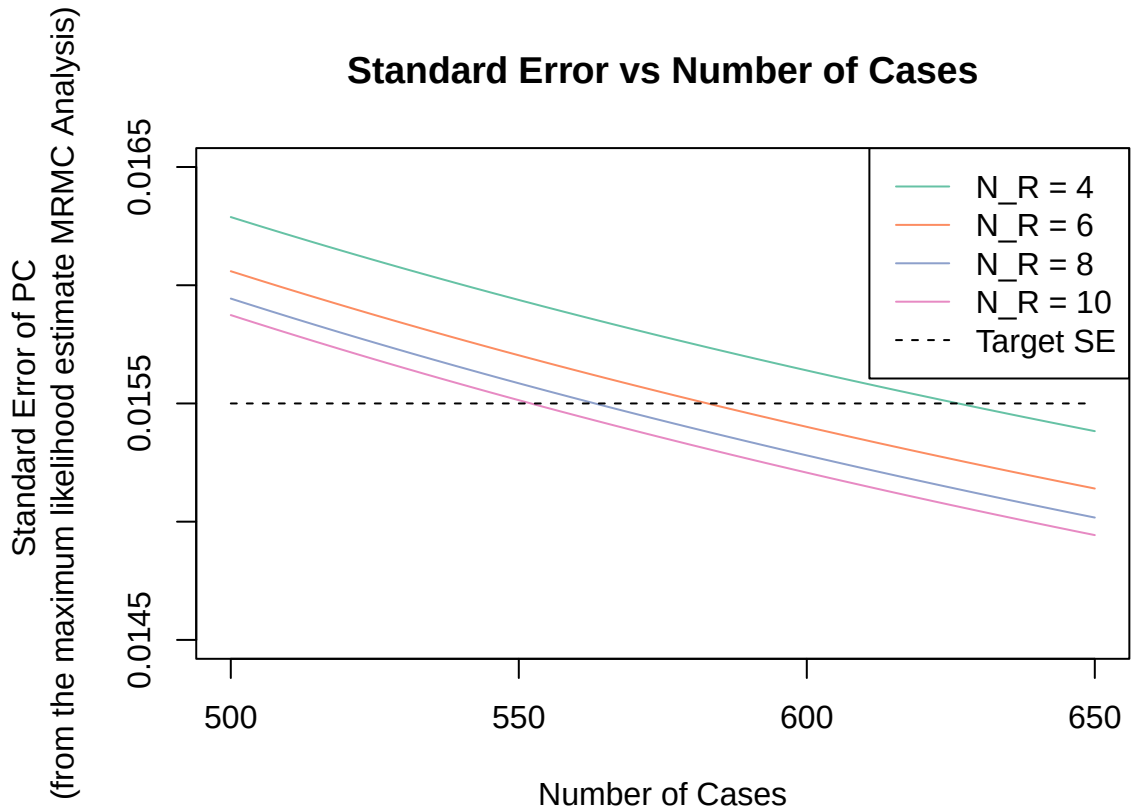
In this example, we consider each case is read by only one reader. Each reader reads a different subset of cases with the same number but no overlap between readers. The coefficients of moments of variance in a 2AFC study with this design are as follows (Gallas, Pennello, and Myers 2007).

c_1	c_4	c_5	c_8
$1/N_C$	$(N_C - N_R)/N_R N_C$	0	$-1/N_R$

By changing the number of cases and the number of readers, we plot the standard error of difference in PCs between DLMO and human readers.

```
# Code is similar to the fully crossed design,
# except the coefficients.

# <... snip ...>
```

```r
c1sp <- 1 / N_C
c4sp <- (N_C - N_R) / (N_R * N_C)
c5sp <- 0
c8sp <- -1 / N_R

# <... snip ...>
```



We observe that standard error with $N_R = 4$ falls below the target standard error after approximately $N_C = 630$. This reveals that *for the considered margin, to show the similarity of DLMO to human readers with a significance level of 0.0125,* ***4*** *readers need to read at least* ***630*** *cases in a flat split-plot 2AFC study without overlap in cases read by readers. This means that each reader needs to read at least 158 cases.*

This study design can be visualized as follow.

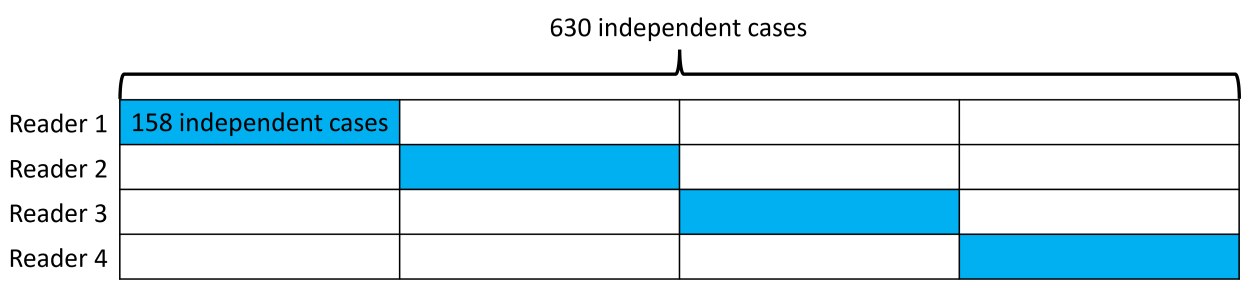


Figure 3: The layout of a flat split-plot design study without overlap between cases read by readers. Blocks colored in blue indicate cases read by readers.

Flat with uniform overlap

Although the above design looks promising, the inter-reader variability cannot be considered and estimated because there was no overlap between cases read by different readers. This may lead to an incomplete analysis. We can design a pivotal study with greater flexibility, where readers are divided into multiple groups and readers in the same group share the same cases. The layout of this design can be visualized in the following figure.

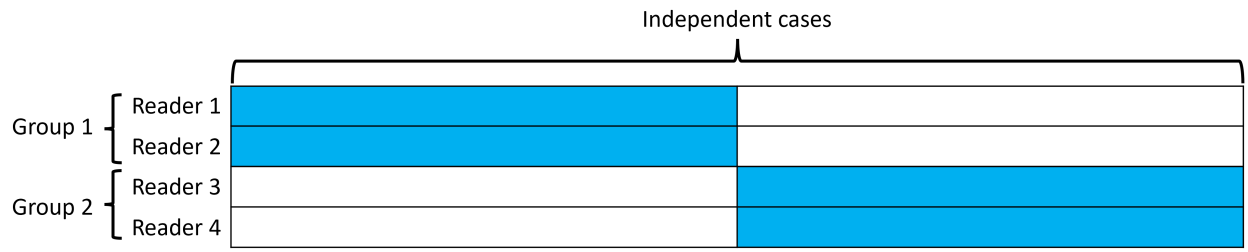


Figure 4: The layout of a flat split-plot design study with uniform overlap. There are two reading groups in this example. Blocks colored in blue indicate cases read by readers.

We denote the number of cases read by the r-th reader by $N_{c|r}$, and denote $w_r = N_{c|r}/N_C$. Also, we define a design matrix with elements denoted by d_{ir}, where i stands for the i-th case and r for the r-th reader. The design matrix holds a one in every position where an outcome was collected and a zero everywhere else. The coefficients of moments of variance are as follows (Gallas, Pennello, and Myers 2007).

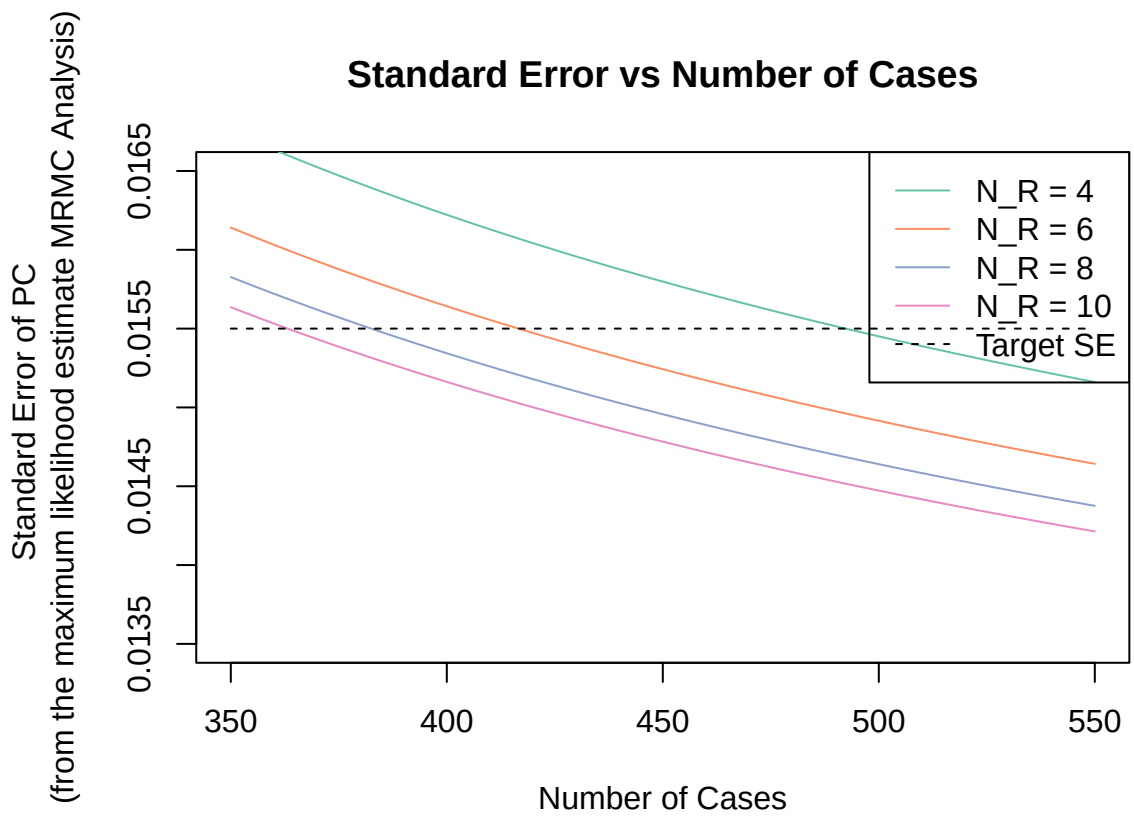
$$
\begin{aligned}
c_1 &= \sum_{r=1}^{N_R} \frac{w_r^2}{N_{c|r}}, \\
c_4 &= \sum_{r=1}^{N_R} w_r^2 - \sum_{r=1}^{N_R} \frac{w_r^2}{N_{c|r}}, \\
c_5 &= \sum_{r=1}^{N_R} \sum_{r' \neq r}^{N_R} \frac{w_r w_{r'}}{N_{c|r} N_{c|r'}} \sum_{i=1}^{N_C} d_{ir} d_{ir'}, \\
c_8 &= \sum_{r=1}^{N_R} \sum_{r' \neq r}^{N_R} w_r w_r' - \sum_{r=1}^{N_R} \sum_{r' \neq r}^{N_R} \frac{w_r w_{r'}}{N_{c|r} N_{c|r'}} \sum_{i=1}^{N_C} d_{ir} d_{ir'} - 1.
\end{aligned}
$$

Since there is overlap between cases read by different readers, coefficient c_5 is no longer zero. In fact, we can estimate the standard error using these coefficients of moments with any design of the pivotal study, whether it is fully crossed, split-plot with or without overlap. For example, we can design a split-plot study with some subset of cases read by all readers (hybrid design).

Fix the number of reading groups We first fix the number of reading groups at two. By changing the number of cases and the number of readers, we plot the standard error of difference in PCs between DLMO and human readers.

```
# Code is similar to the fully crossed design,
# except the coefficients.

# <... snip ...>

# Number of reading groups
N_g <- 2
```

```r
# <... snip ...>

N_c_r <- N_C / N_g # number of cases per reader
w_r <- N_c_r / (N_c_r*N_R) # weight
N_c_r_overlap <- N_c_r # number of overlapped cases per reader

c1sp <- N_R * w_r^2 / N_c_r
c4sp <- N_R * w_r^2 - c1sp
c5sp <- N_R * w_r^2 / N_c_r^2 * N_c_r_overlap * (N_R/N_g - 1)
c8sp <- w_r^2 * N_R * (N_R-1) - c5sp - 1

# <... snip ...>
```



We observe that standard error with $N_R = 4$ falls below the target standard error after approximately $N_C = 500$. This reveals that *for the considered margin, to show the similarity of DLMO to human readers with a significance level of 0.0125,* ***4*** *readers need to read at least* ***500*** *cases in a flat split-plot 2AFC study with* ***two*** *reading groups. Readers in the same reading group read the same set of images. This means that each reader needs to read at least 250 cases.*

Fix the number of readers In many situations, investigators have limited flexibility in the number of readers they can recruit, so they have to pay more attention to the reader assignment, for example, by changing the number of reading groups in a split-plot design study. Of course, if there is only one reading group, this design is identical to the fully crossed design. In the following analysis, we demonstrate the effect of the number of reading groups on the standard error with a fixed number of readers. Since we need readers to be evenly assigned among sessions and 12 can be divided by many numbers, we consider 12 readers here

for the demonstration purposes. An example of this design with four reading groups is shown here.

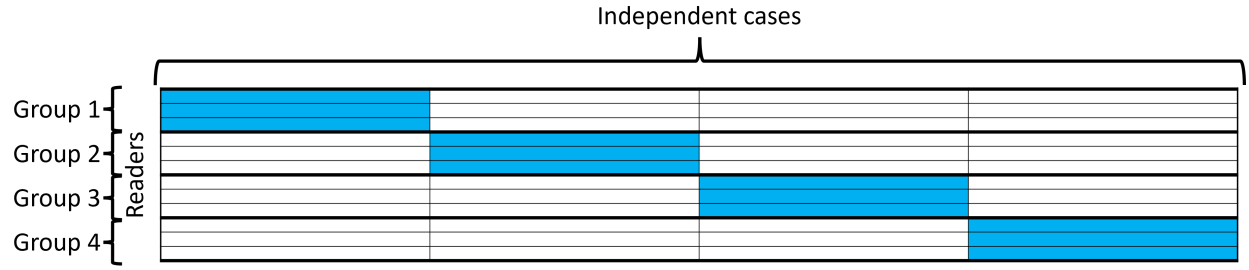
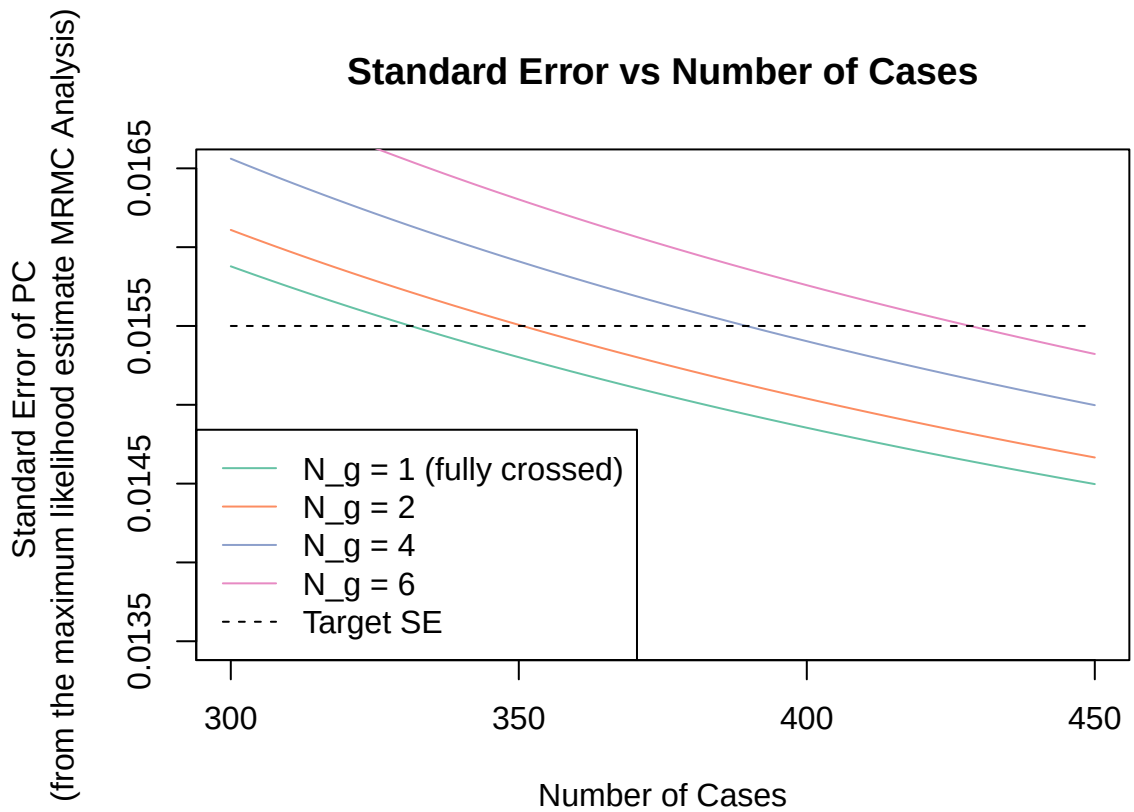


Figure 5: The layout of a flat split-plot design study with uniform overlap. There are four reading groups and 12 readers in this example. Blocks colored in blue indicate cases read by readers.

By changing the number of cases and the number of reading groups, we plot the standard error of difference in PCs between DLMO and human readers.

```
# Code is almost similar to the fully crossed design.

# <... snip ...>

# Set N_R range
N_g_start <- 2
N_g_end <- 6

# Number of readers
N_R <- 12

# <... snip ...>

for (N_g in c(1,seq(N_g_start, N_g_end, by = 2))){

  # <... snip ...>

  N_c_r <- N_C / N_g # number of cases per reader
  w_r <- N_c_r / (N_c_r*N_R) # weight
  N_c_r_overlap <- N_c_r # number of overlapped cases per reader

  c1sp <- N_R * w_r^2 / N_c_r
  c4sp <- N_R * w_r^2 - c1sp
  c5sp <- N_R * w_r^2 / N_c_r^2 * N_c_r_overlap * (N_R/N_g - 1)
  c8sp <- w_r^2 * N_R * (N_R-1) - c5sp - 1

  # <... snip ...>

}

# <... snip ...>
```



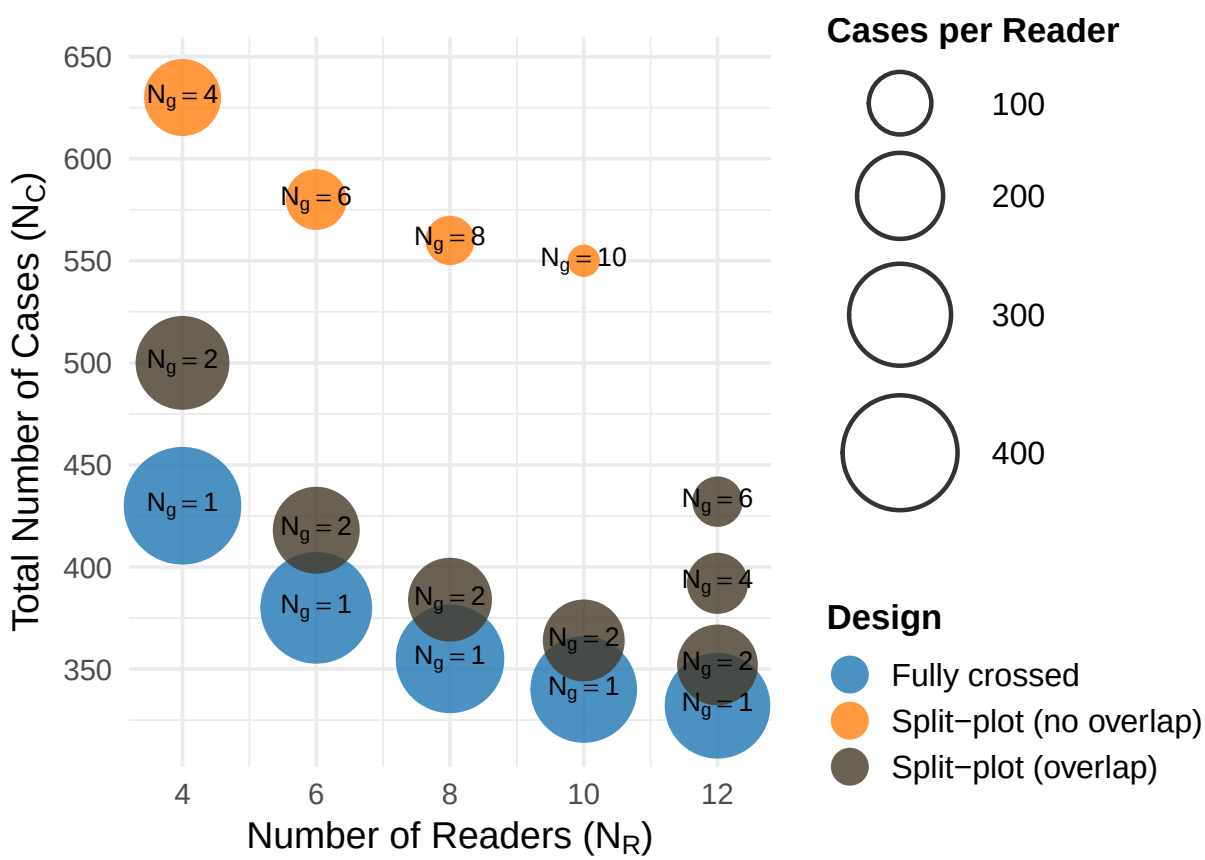
We observe that standard error with $N_g = 6$ falls below the target standard error after approximately $N_C = 432$. Each reading group has two readers. This reveals that *for the considered margin, to show the similarity of DLMO to human readers with a significance level of 0.0125,* ***12*** *readers need to read at least* ***432*** *cases in a flat split-plot 2AFC study with* ***six*** *reading groups. Readers in the same reading group read the same set of images. This means that each reader needs to read at least 72 cases.*

Discussion and Conclusion

The fully crossed and split-plot designs are summarized below.

Design	N_g	N_R	N_C	Number of cases per reader	Estimated standard error
Fully crossed	1	4	430	430	0.015466
Fully crossed	1	6	380	380	0.015455
Fully crossed	1	8	355	355	0.015474
Fully crossed	1	10	340	340	0.015495
Fully crossed	1	12	332	332	0.015492
Split-plot (no overlap)	4	4	630	158	0.015481
Split-plot (no overlap)	6	6	580	97	0.015495
Split-plot (no overlap)	8	8	560	70	0.015495
Split-plot (no overlap)	10	10	550	55	0.015495

Design	N_g	N_R	N_C	Number of cases per reader	Estimated standard error
Split-plot (overlap)	2	4	500	250	0.015451
Split-plot (overlap)	2	6	418	209	0.015488
Split-plot (overlap)	2	8	384	192	0.015486
Split-plot (overlap)	2	10	364	182	0.015492
Split-plot (overlap)	2	12	352	176	0.015486
Split-plot (overlap)	4	12	392	98	0.015496
Split-plot (overlap)	6	12	432	72	0.015469



As the number of readers increases, the number of cases required decreases in both fully crossed and split-plot designs (if the number of reading groups is fixed or readers read no overlap cases). The fully crossed design achieves a similar estimated standard error with fewer cases than the split-plot design. On the other hand, although the split-plot design requires more cases than the fully crossed design, the number of cases read by each reader is lower than that in the fully crossed design.

The final choice of pivotal study design depends on the available resources. These resources include the number of cases that can be obtained, the number of readers that can be recruited, and the amount of time readers can contribute or associated cost if the readers must be paid (which relates to the number of cases per

reader). If cases are more limited, the fully crossed design can be chosen because it requires fewer cases, and when reader time or cost is constrained, the split-plot design can be selected. However, the design selection really depends on the specific study context. One effective strategy is to first fix one of these elements and then explore the others.

Because images are synthetic in this study, cases can be generated as needed. However, we can only recruit a few readers ($N_R = 4$), and minimizing their time commitment is desirable given their existing workload. Thus, the primary constraint is the readers' time availability. The number of images read per reader is substantially lower in the split-plot design without overlap compared with other designs. Therefore, we selected this design with $N_R = 4$ and $N_C = 640$, requiring each reader to evaluate $640/4 = 160$ trials in the pivotal study.

References

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